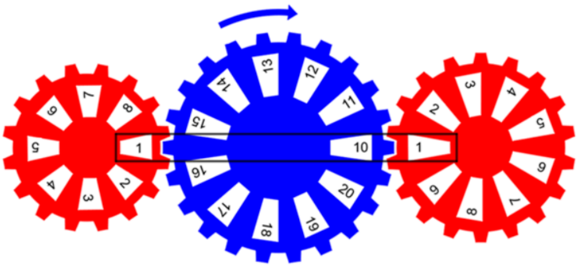
S1Q01

Question 1 - Stage 1



As the cogs rotate, with the central cog always going clockwise, Ada adds together all 792 different 4-digit numbers she sees in the rectangle, starting with 1101 as shown. Her sum is S.

Will, on the other hand, multiplies the three numbers in the rectangle together. The sum of these 792 products is T.

What is the value of $\frac{S}{T}$ (to 4 s.f.)?

Code

In [1]:

```python
from itertools import product
from pyTools import roundSF
from math import prod


cog0 = ["1","2","3","4","5","6","7","8"]
cog1 = ["10","11","12","13","14","15","16","17","18","19","20"]
cog2 = ["1","2","3","4","5","6","7","8","9"]

sumTotal = 0
productTotal = 0

for gearPosition in product(cog0,cog1,cog2):
    sumTotal += int("".join(gearPosition))
    productTotal += (prod(int(x) for x in gearPosition))

print(f"Sum Total: {sumTotal}")
print(f"Product Total: {productTotal}")
print(f"S/T: {roundSF(sumTotal/productTotal,4)}")
```

```
Sum Total: 3686760
Product Total: 267300
S/T: 13.79
```

Pure Maths

$$
\begin{aligned}
&C_0 \in \{1,2,3,4,5,6,7,8\} \\
&C_1 \in \{10,11,12,13,14,15,16,17,18,19,20\} \\
&C_2 \in \{1,2,3,4,5,6,7,8,9\}
\end{aligned}
$$

$$
\begin{aligned}
&\text{let } s = C_0 + C_1 + C_2 \\
&\text{let } t = C_0 \cdot C_1 \cdot C_2
\end{aligned}
$$

$$
\frac{S}{T} = \frac{\frac{S}{792}}{\frac{T}{792}} = \frac{\bar{s}}{\bar{t}}
$$

$$
\begin{aligned}
\bar{s} &= 1000 \cdot \bar{C_0} + 10 \cdot \bar{C_1} + \bar{C_2} \\
&= 1000 \cdot 4.5 + 10 \cdot 15 + 5 \\
&= 4655
\end{aligned}
$$

$$
\begin{aligned}
\bar{t} &= \bar{C_0} \cdot \bar{C_1} \cdot \bar{C_2} \\
&= 4.5 \cdot 15 \cdot 5 \\
&= 337.5
\end{aligned}
$$

$$
\begin{aligned}
\frac{\bar{s}}{\bar{t}} &= \frac{4655}{337.5} \\
&= 13.7925\ldots \\
&\approx 13.79
\end{aligned}
$$